Seventeen further literal structure labels in the M_{11} table

L054 bounded recovery component

Acquired interface and source. We acquire the accepted twenty-four-label pilot's elementary witness rules, explicit transport proposition, actual subgroup arrays, numerical row maps and native interpretation of Connor–Leemans' extension notation. The source is the current retained two-page companion M_{11} table; its historical identity with every fixed-2013 file remains unasserted. Section 3 of the final 2015 journal article supplies the notation: a colon is a semidirect product, a centered dot is a nonsplit extension, and unparenthesized products associate to the left. Native pages 1 and 2, including their continued-table boundary, were inspected directly. No flattened numeral or punctuation heuristics determine the following label trees.

Theorem 1. *The actual saved subgroup representatives joined to the following seventeen source rows have the displayed structures:*

row	*structure*	*row*	*structure*
8	$((3 : S_3) \cdot 2) \cdot 2$	9	$((3 : S_3) \cdot 2) : 2$
10	A_5	11	A_5
15	$(3 : S_3) \cdot 2$	23	D_{12}
24	A_4	25	11
26	D_{10}	31	S_3
32	S_3	33	6
34	5	35	4
37	3	38	2
39	1		

Bare integers denote cyclic groups, and a dihedral subscript denotes order.

Proof. The exact original M11 generator list is parsed by the acquired literal token parser. Each new row has its previously accepted numerical row index and an exact generator-array join; neither an order coincidence nor a structural name is used to substitute a subgroup. The verifier computes each finite group's complete literal elements from those generators and checks the certificate against the manually transcribed native tree.

All atom, normality, intersection, complement, generation and multiplication tests use the unchanged accepted pilot verifier. In row 8 the first normal factor of order 36 is itself certified as $(3 : S_3) \cdot 2$, and its embedding is normal of index two in the actual order-72 group. Both index-two quotient maps are nonsplit: every outside element is checked and none has order two. Row 9 has an actual outer complement, while the inner order-36 factor is a nonsplit index-two extension: all eighteen outside elements have order different from two. These are distinct positive assertions about the displayed left-associated trees; no uniqueness of descriptions is inferred. The remaining rows use exactly the acquired cyclic, alternating, symmetric, dihedral and index-two rules. Thus the new witnesses pay twenty-nine factor nodes, 7,524 standard-model multiplication checks and ninety exhaustive outside-element checks.

The registry is extended by retaining all twenty-four old concrete models and all forty-four old occurrence maps as literal prefixes. Five new concrete models suffice for the twenty-nine new factor occurrences. Every new map is an explicit bijective homomorphism, checked on every product using the acquired transport verifier. The resulting twenty-nine concrete models serve seventy-three factor occurrences, with 38,910 checked transport products. In particular, the two displayed A_5 rows may share a model without identifying their ambient conjugacy classes. All hypotheses of the acquired certificate and transport propositions are thereby met. □

Execution and scope. The discovery was capped at 120 seconds and finished in less than one second, using only the seventeen retained subgroups of order at most 72. No ambient census was run. The final checker performs no discovery or isomorphism search; it verifies literal witnesses and rejects ten damaged joins, nested extension witnesses and transport maps, also with Python optimization enabled. Together with the accepted pilot this accounts for the nine A_5 and thirty-two M_{11} labels. M11 rows 1–7, involving $M_{11}, M_{10}, L_2(11), A_6, M_9 : 2, S_5, M_9$, remain outside this component. No canonical name, historical naming-software reproduction, whole original, conjecture, Lean or measurement claim is made. The current author supplied the extension and its code; root proposed registry reuse and authorized the fixed roster. The predecessor's row 9 AST wrongly transcribed the first centered dot as a colon; root review caught this source error. This successor certifies the actual printed nonsplit-inner, split-outer label and checks AST punctuation against the literal source. The other sixteen witnesses remain byte-identical. Distinct review remains required.